Chapter 5: Eigenvalues and Eigenvectors

Eigenvalues are one of the most important things you learn in this class. Both because they come up in many contexts in this course but also in many other classes, data science couses for instance, they are common.

You will see them quite often in the later linear algebra content, Chapters 6 and 7. And perhaps surprisingly they will even come up in the differential equations part of the class! So definitely when you prepare for the Final Exam, eigenvalues should be one of the top things on your list to study / try to master.

5.1 Introduction

One use relates to following dilemma. Not the most important use perhaps, but maybe the easiest one to understand the motivation behind why we might need help to solve the dilemma in a reasonable manner.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{100} = ?$$

This 2 x 2 matrix is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$. Maybe something like stretching and rotating put together?

Whatever it is precisely, if we want to apply it 100 times how to do so? Remember this is a linear transformation where the domain and codomain are the same (square matrix) so it makes sense to try to apply it many times in succession and see what happens.

But how to do that in an efficient manner is the big question. Certainly can't do that manually. In contrast, note for this other matrix we have

$$\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}^{100} = \begin{bmatrix} 1 & 0 \\ 0 & 4^{100} \end{bmatrix}$$

That is very easy to do since diagonal matrix - this is no problem for that reason . Can we use that idea in the earlier problem even though not diagonal matrix?

Not directly perhaps, but maybe with help of eigenvalues it is feasible. Speculative at moment, but that's one of many things eigenvalues help with.

Related is that diagonal matrices are much easier to work with so this becomes a consistent dilemma, "converting" matrices to diagonal to make them easier to work with.

We actually use that a lot in rest of class, even in the differential equations part of the class (hence part of the reason for the crossover).

For an $n \times n$ matrix A, λ is an <u>eigenvalue</u> for A if there is a nonzero vector $x \in \mathbb{R}^n$, for which

$$\underbrace{A}_{n \times n} \underbrace{x}_{n \times 1} = \underbrace{\lambda}_{\text{scalar}} \underbrace{x}_{n \times 1}$$

That is, A maps x to a constant multiple of itself. Both sides are $n \times 1$ so this does make sense, but of course being same size doesn't guarantee they are equal. Likely a very limited set of λ's that such x exist for.

This is the definition of eigenvalue, and can use it for general linear transformations, from any vector space to itself, i.e. doesn't have to be from $\mathbb{R}^n$ to $\mathbb{R}^n$ or even have to be a matrix transformation (could define these for linear transformations from an infinite dimensional vector space to itself. Though certainly with matrices, i.e. from $\mathbb{R}^n$ to $\mathbb{R}^n$ would be the most common context in this course.

One interesting thing to note is that from $Ax = \lambda x$ we get that

$$A^2 x = A(Ax) = A(\lambda x) = \lambda (Ax) = \lambda^2 x$$

Continuing like this, we can get that:

$$A^{100} x = \lambda^{100} x$$

So evaluating the 100th power of the matrix at any eigenvector is fairly straightforward. Hence not surprising that those might help us compute the 100th power of the matrix in general.

Though $Ax = \lambda x$ is the way we "define" eigenvalues that doesn't tend to be the way we look at eigenvalues or how we try to actually find them.

The more common thing for that is:

$$(A - \lambda I)x = 0$$

That is, for which λ does $A - \lambda I$ have a nontrivial null space (hence last chapter plays big role here). Also since $A - \lambda I$ is square can associate this with whether or not it is invertible.

If such a nontrivial x exists for a certain λ, then that x is an <u>eigenvector</u> for A associated with λ.

And the set of all such x's for such a λ together with the 0 vector are the <u>eigenspace</u> of A associated with λ. And this is a subspace because it contains the 0 vector and since each x in it satisfies $Ax = \lambda x$ we can tell quickly that this set is closed under addition and scalar multiplication.

Note that the 0 vector is in any eigenspace, for one since that set would not be subspace without it and because it does always satisfy $Ax = \lambda x$, no matter the λ .

That said, 0 is never considered to be an eigenvector because of that last issue. That is, when searching for eigenvalues, 0 always satisfies $Ax = \lambda x$ so if 0 were permitted to be an eigenvector, every λ would always be an eigenvalue which is not the point.

5.1.7

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$$, is $\lambda = 4$ an eigenvalue?

Let's check if matrix $A - 4I$ has nontrivial null space.

$$A - 4I = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix} \quad \text{----->} \quad \begin{array}{c} \text{Replace:} \\ \text{R2 with R2 + 2R1} \\ \text{R3 with R3 - 3R1} \end{array}$$

We could check determinant, but if we also wanted eigenvectors (which we usually will), row reduction will help us find those as well (after all it's about null space) so that's the better general approach.

$$\sim \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \\ 0 & 4 & 4 \end{bmatrix} \quad \text{----->} \quad \begin{array}{c} \text{Replace:} \\ \text{R3 with R3 + 4R2} \end{array}$$

$$\sim \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Now in REF. Last column nonpivot so check null space:

$$-x_2 - x_3 = 0$$
$$-x_1 - x_3 = 0$$

Since x_3 only free variable, null space is:

$$x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \text{ ----> } \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \text{ : eigenvector}$$

That's mostly how to check if some particular value is an eigenvalue, i.e. check if $A - \lambda I$ has a nontrivial null space in whatever way is convenient.

But how to find these prospective eigenvalues to begin with? Certainly can't test values one by one.

Well oftentimes when dealing with a tough matrix we did row reduction to make it easier to work with. Might that be an option here?

Well have to be careful. Row operations do not change row or null space but do potentially change say column space. So what about eigenvalues/eigenvectors?

Well when talking about determinants, we first figured out how triangular matrices worked and then expanded. So will do same here.

For triangular matrices, eigenvalues are just the diagonal entries? Why? Well subtract any of them from full diagonal, matrix is still triangular with at least one 0 on the diagonal so has determinant 0 which is the key.

So that's helpful, but it also immediately tells you that row operations will affect eigenvalues as you say scale a row of a triangular matrix, that will usually change its diagonal entry, thereby altering that associated eigenvalue (i.e. whenever it's nonzero).

Additionally note:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{vs.} \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

These are quite clearly row equivalent. But note that

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

So $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is an eigenvector for first matrix, does map to multiple of itself. On other hand,

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Meaning $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is not an eigenvector of this matrix.

Thus eigenvectors, without worrying about the context of associated eigenvalues, just thinking of those as nonzero vectors which the matrix maps to a multiple of themselves can be altered via row operations too.

Thus row operations do potentially change eigenvalues/eigenvectors so row reduction is not generally an approach to find either.

This might surprise some as we said eigenvalues relate to null space and row operations don't affect null space.

Key is: we need null space of $A - \lambda I$ NOT A.

So the big problem is if you row reduce A that won't reduce the λI portion as would be needed. That's why row operations do usually change eigenvalues.

So if you want to do row reduction would really have to row reduce $A - \lambda I$. Can you do that to find eigenvalues? Yes but is it desirable?

Thing is the presence of the λ complicates that.

Trying to change into REF, might have to manipulate with the λ involved which is much more annoying, you saw some problems early in course with such unknowns. Also have to worry about multiplying or dividing by 0 due to the unknown element.

So can be done but usually not preferred. But one thing we do have is that even though we are interested in null space, matrix is square so we can use determinants to maybe make things easier.

That's largely what next section is about.

Before that: very important result is that in a matrix A a set of eigenvectors which correspond to distinct eigenvalues is always linearly independent. Why?

Say $\{v_1, v_2, \ldots, v_p\}$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$ and are linearly independent.

Now say we add a new eigenvector v_{p+1} to the set which corresponds to a new eigenvalue λ_{p+1}
And say the set of vectors now is linearly dependent.

Then the new eigenvector must be a linear combination of the original ones meaning we must have

$$c_1 v_1 + c_2 v_2 + \ldots + c_p v_p = v_{p+1}$$

---> $$A[c_1 v_1 + c_2 v_2 + \ldots + c_p v_p] = A v_{p+1}$$

---> $$c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \ldots + c_p \lambda_p v_p = \lambda_{p+1} v_{p+1}$$

based on these being eigenvalues/eigenvectors.

Now λ_{p+1} times top minus the bottom yields

$$c_1(\lambda_{p+1}-\lambda_1)v_1 + \ldots + c_p(\lambda_{p+1}-\lambda_p)v_p = 0$$

So since $\{v_1, v_2, \ldots, v_p\}$ is linearly independent this forces:

$$c_1(\lambda_{p+1}-\lambda_1) = c_2(\lambda_{p+1}-\lambda_2) = \ldots = c_p(\lambda_{p+1}-\lambda_p) = 0$$

But since λ_{p+1} distinct from prior eigenvalues we must then have

$$c_1 = c_2 = \ldots = c_p = 0$$

which forces $v_{p+1} = 0$, contradiction since eigenvector can't be 0.

Thus set stays linearly independent. Hence starting with one vector (nonzero vector always linearly independent on its own) and adding one new eigenvector at a time (each corresponding to a new eigenvalue), inductively such a set of eigenvectors is always linearly independent.

Example of finding eigenvectors/eigenspace. Very important thing to be able to do in many contexts.

5.1.13

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \qquad \lambda = 1, 2, 3$$

Here they give you the eigenvalues which won't be the usual thing, most of the time you have to find them yourself.

Here they do so for a couple reasons probably, one we haven't really learned how to find them (that's next section) and because finding them for 3 x 3 and larger can be tough/long if there isn't some trick.

But longterm, 3 x 3 will be the common setting where you will be expected to find eigenvalues from scratch.

$\lambda = 1$:

$$A - I = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

null space?

$$3x_1 + x_3 = 0 \text{ ----> } x_3 = 0$$
$$-2x_1 = 0 \text{ ----> } x_1 = 0$$

$x_1 = x_3 = 0$, x_2 free so eigenvector here is $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

So that generates that eigenspace (only one dimension since one free variable).

$\lambda = 2$:

$$A - 2I = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

null space?

$$2x_1 + x_3 = 0$$
$$2x_1 + x_2 = 0$$

Pick $x_1 = 1$ then $x_2 = x_3 = -2$, thus eigenvector is $\begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$

Again only one free variable (A - 2I clearly has rank two based on simplified form), so that generates the eigenspace.

$\lambda = 3$:

$$A - 3I = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

null space?

$$x_1 + x_3 = 0$$
$$x_1 + x_2 = 0$$

Pick $x_1 = 1$ then $x_2 = x_3 = -1$, thus eigenvector is $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

Again, dimension one only since $A - 3I$ has two pivots.

All of these eigenspaces had dimension one. Could it be possible for dimension to be higher? If so, under what circumstances? That actually is a super important issue in this context/chapter. One of several things we will try to clarify in next sections.

5.2 Characteristic Equation

So far we have been provided eigenvalues for the matrices which we solved. But how to find from scratch?

Again row reduction maybe not desirable. Usual way is to check equation

$$\det(A - \lambda I) = 0$$

as solving that for λ will give us the eigenvalues of A. We call $\det(A - \lambda I)$ the <u>characteristic polynomial</u> of A, and $\det(A - \lambda I) = 0$ is the <u>characteristic equation</u> for A.

5.2.13

$$A = \begin{bmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{bmatrix}$$

Want eigenvalues so let's set up the characteristic equation:

$$A - \lambda I = \begin{bmatrix} 6-\lambda & -2 & 0 \\ -2 & 9-\lambda & 0 \\ 5 & 8 & 3-\lambda \end{bmatrix}$$

Expanding in third column gives:

$$\det(A - \lambda I) = (3-\lambda)\begin{vmatrix} 6-\lambda & -2 \\ -2 & 9-\lambda \end{vmatrix}$$

$$= (3-\lambda)[(6-\lambda)(9-\lambda) - 4]$$
$$= (3-\lambda)(\lambda^2 - 15\lambda + 50)$$
$$= (3-\lambda)(\lambda-5)(\lambda-10)$$

So setting this to 0 we find

$$\lambda = 3, 5, 10 \quad \text{eigenvalues}$$

Even though the λ complicates things, general ideas same. In particular trying to expand in row or column with as many 0s as possible still holds.

This is kind of an illustration of how "cumulative" this chapter is. You need to know about determinants for sure from Chapter 3. Also about null space from Chapter 4. And the ideas using null space often rely on parametric vector form from Chapter 1.

So a good deal of older things required - you can't really be adept at this new content if you don't have a good grasp of the older content.

Same strategy no matter the size of the matrix, just as usual the bigger the matrix, the harder the calculations.

Even for 3 x 3 this gets hard. In the prior situation, one eigenvalue was apparent due to the column with all but one 0. For a "generic" 3 x 3 you in theory would have to solve a generic cubic equation.

Since not really reasonable to do that, usually has to be some trick, even for that size.

But ok, lot of questions to clarify in this. Since we wind up with quadratics, what if they have nonreal roots?

Also, what if there are repeated roots? How does that affect dimension of eigenspace? Lot that needs clarification.